Spatial DBs

[almost EVERYTHING has a 'spatial aspect'!]

TOC

- spatial DBs - intro, use cases etc.
- spatial datatypes
- spatial DB architecture
- spatial functionality, querying
- indexing
- implementations
- visualization
- more

What is spatial data, what is a spatial database?

"A spatial database is a database that is optimized to store and query **SPATIAL data**, ie. data that can be REPRESENTED using points, lines and polygons (and pixels), ie. using spatial primitives/types."

In other words, it includes data that has a SPATIAL location (and extent).

A chief category of spatial data is **geospatial** data - derived from the geography of our earth.

Characteristics of geographic data:

- has location
- has size
- is usually auto-correlated
- scale dependent
- might be temporally dependent too

Geographic data is NOT 'business as usual'!

Entity view vs field view

In spatial data analysis, we distinguish between two conceptions of space:

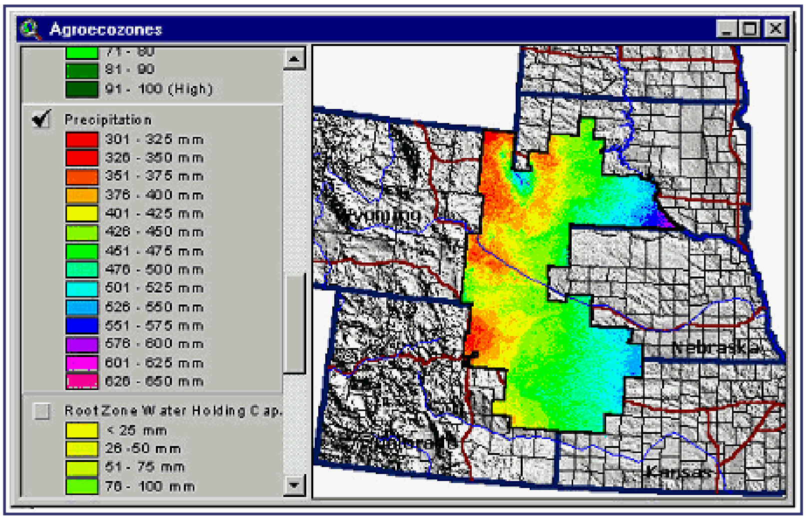
- entity view: space as an area filled with a set of discrete objects
- field view: space as an area covered with essentially continuous surfaces

For our (spatial data) purposes, we will adopt the 'entity' view, where space is populated by discrete objects (roads, buildings, rivers..).

A few examples of spatial data

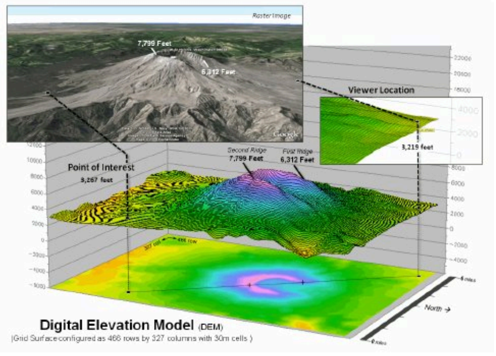
CAD data:

Agricultural data:



3D data:

- Three-dimensional data examples
  - Weather
  - Cartesian coordinates (3-D)
  - Topological
  - Satellite images



More:

- **Army Field Commander**: Has there been any significant enemy troop movement since last night?
- **Insurance Risk Manager**: Which homes are most likely to be affected in the next great flood on the Mississippi?
- **Medical Doctor**: Based on this patient's MRI, have we treated somebody with a similar condition ?
- **Molecular Biologist**:Is the topology of the amino acid biosynthesis gene in the genome found in any other sequence feature map in the database ?
- **Astronomer**:Find all blue galaxies within 2 arcmin of quasars.

Examples of geospatial data

- spread of disease, risk of disease [look at this too]
- drug overdoses - over time
- census data
- income distribution, home prices
- locations of Starbucks (!)
- (real-time) traffic
- agricultural land use, deforestation
- crime data [look at these maps, overlaid with crime data... (eg. LAPD Online, and ArcGIS Online)]

Various government agencies routinely coordinate spatial data collection and use, operating in effect, a national spatial data infrastructure (NSDI) - these include federal, state and local agencies. At the federal level, participating agencies include:

- Department of Commerce
  - Bureau of the Census
  - NIST
  - NOAA
- Department of Defense
  - Army Corps of Engineers
  - Defense Mapping Agency
- Department of the Interior
  - Bureau of Land Management
  - Fish and Wildlife Service
  - U.S Geological Survey [earthquakes, map projections]
- Department of Agriculture
  - Agricultural Stabilization and Conservation Service
  - Economic Research Service
  - Forest Service
  - National Agriculture Statistical Service
  - Soil Conservation Service
- Department of Transportation
  - Federal Highway Administration
- Environmental Protection Agency
- NASA

As you can see, spatial data is a SERIOUS resource, vital to US' national interests.

Where does geo/spatial data come from?

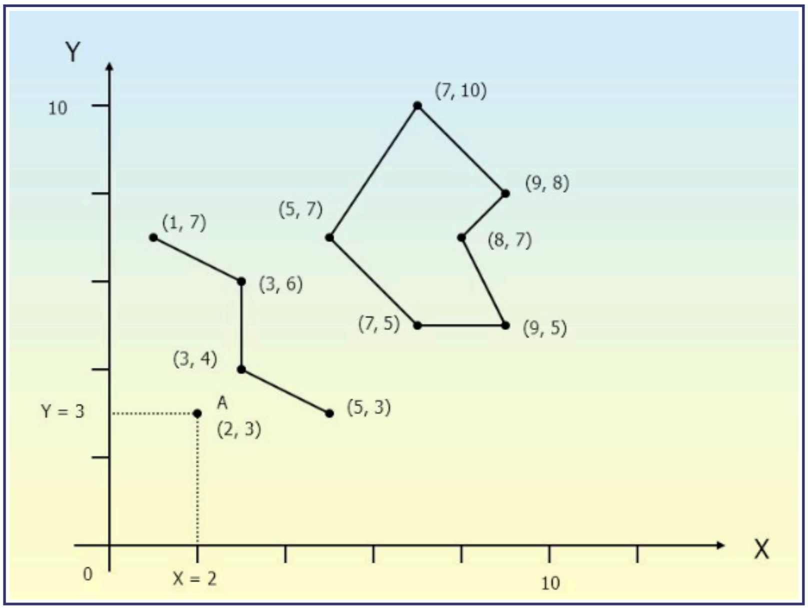
Spatial data is created in a variety of ways:

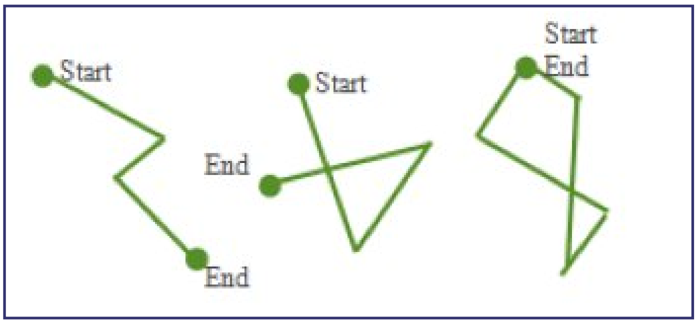
- CAD: user creation
- CAD: reverse engineering
- maps: cartography (surveying, plotting)
- maps: satellite imagery
- maps: 'copter, drone imagery
- maps: driving around
- maps: walking around

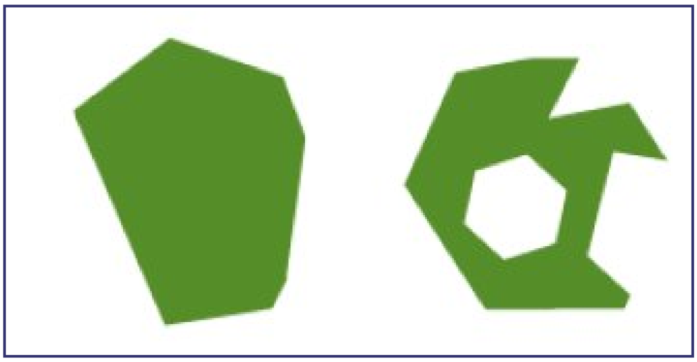
What to store?

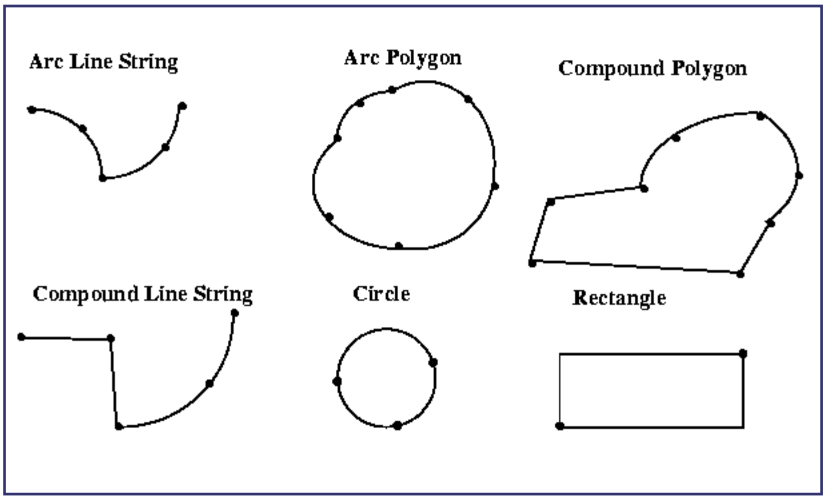
All spatial data can be described via the following entities/types:

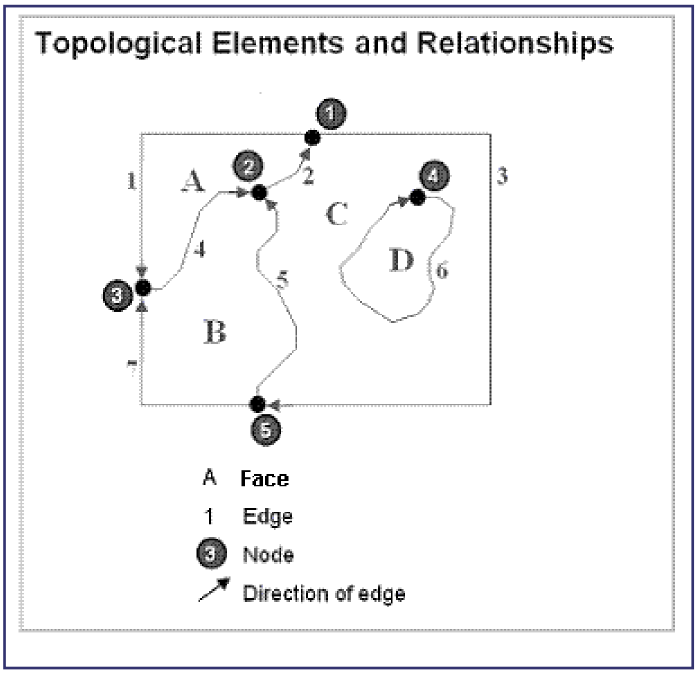
- points/vertices/nodes
- polylines/arcs/linestrings
- polygons/regions
- pixels/raster

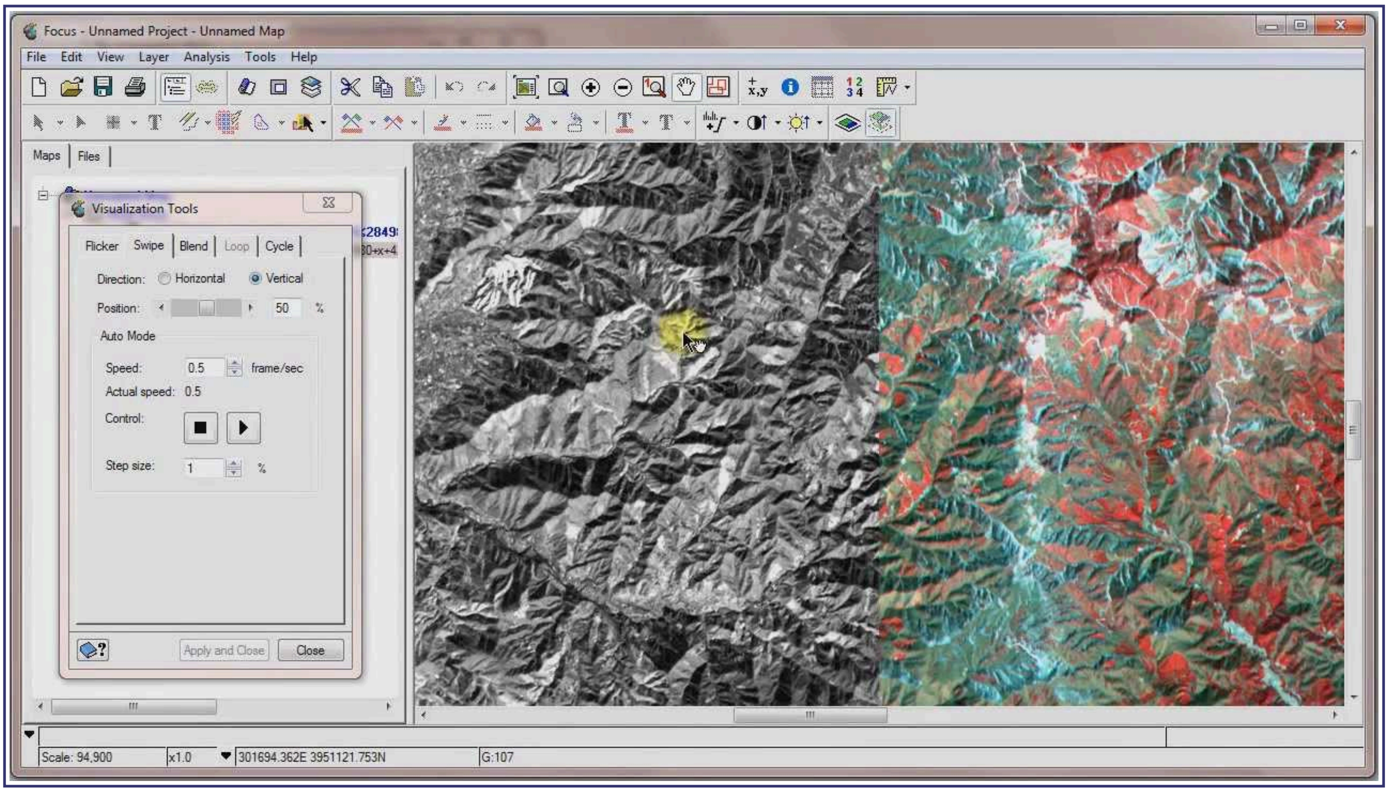






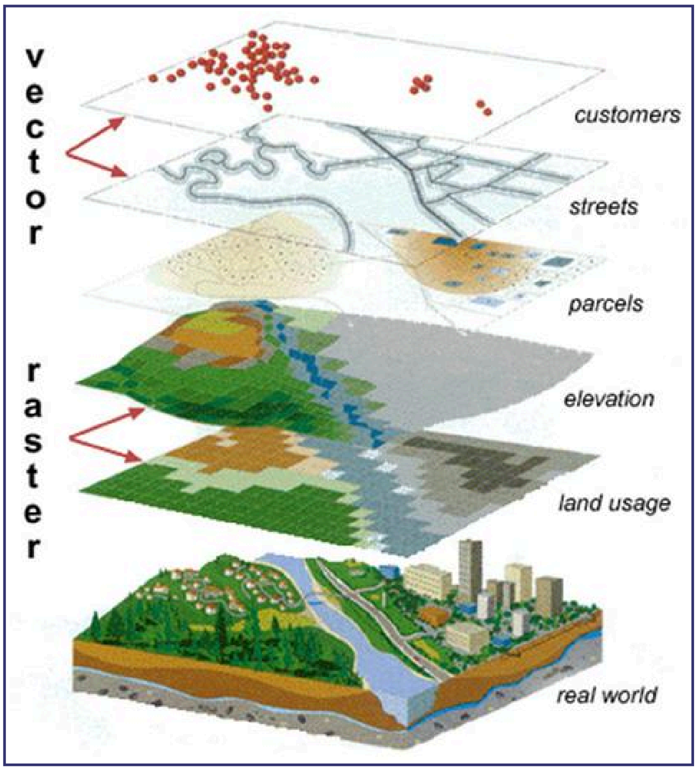

Arc Line String
Arc Polygon
Compound Polygon
Compound Line String
Circle
Rectangle

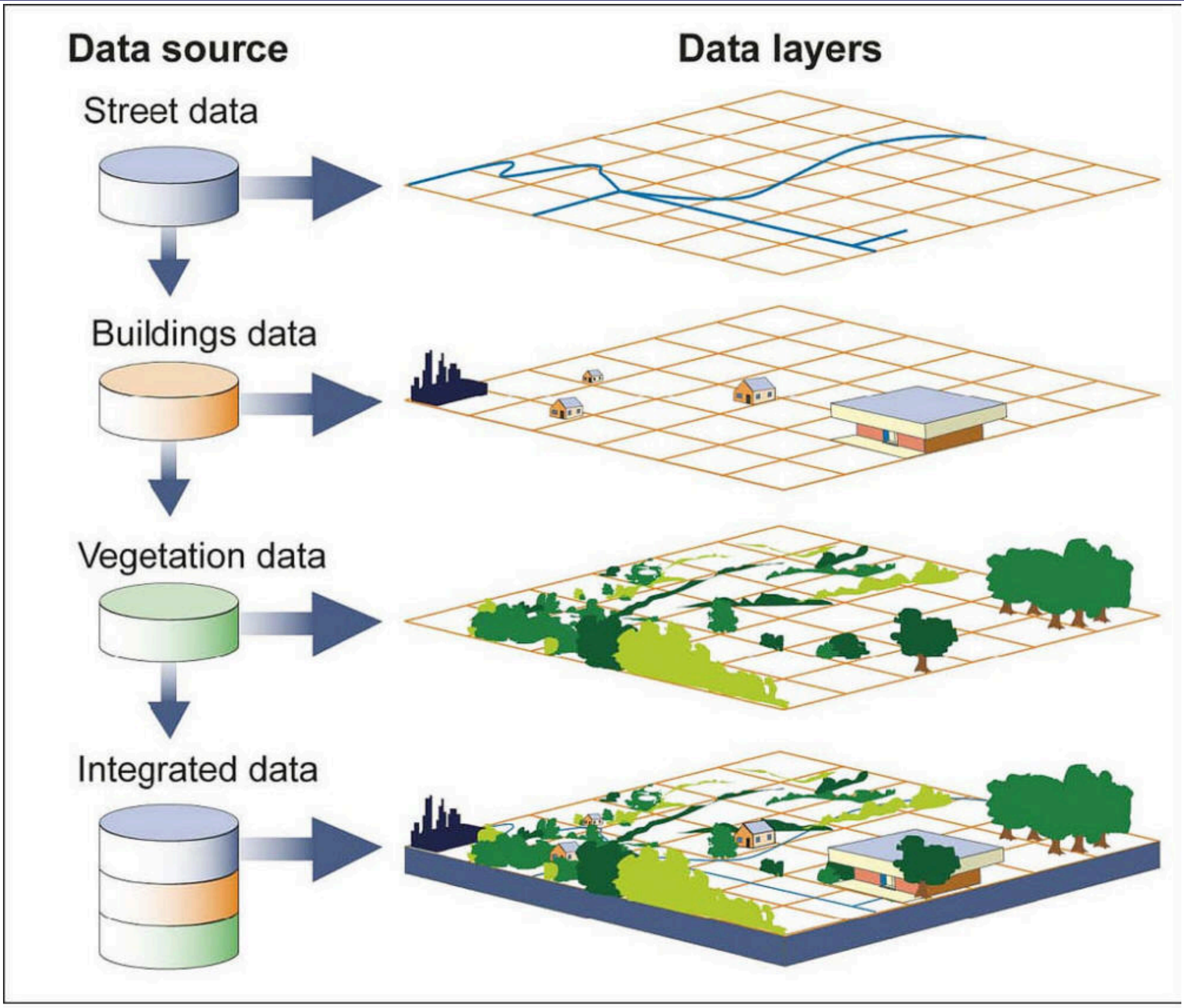

Topological Elements and Relationships
A
B
C
D
1
2
3
4
5
6
7
A Face
1 Edge
Node
Direction of edge


Focus - Unnamed Project - Unnamed Map
File Edit View Layer Analysis Tools Help
Maps Files
Visualization Tools
Flicker Swipe Blend Loop Cycle
Direction: Horizontal Vertical
Position: 50 %
Auto Mode
Speed: 0.5 frame/sec
Actual speed: 0.5
Control:
Step size: 1 %
Apply and Close Close
Scale: 94,900
x1.0
301694.362E 3951121.753N
G:107

Once we have spatial data (points, lines, polygons), we can:

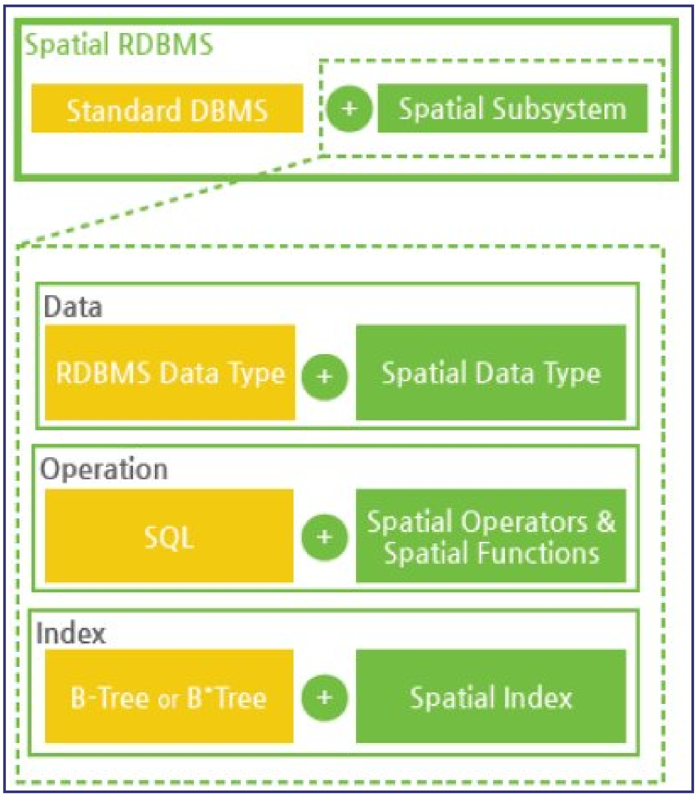
- 'model' features such as lakes, soil type, highways, buildings etc, using the geometric primitives as underlying types
- **add 'extra', non-spatial attributes/features to the underlying spatial data**




Data source
Data layers
Street data
Buildings data
Vegetation data
Integrated data

Source: GAO.

Components of a spatial DB, apps built over it

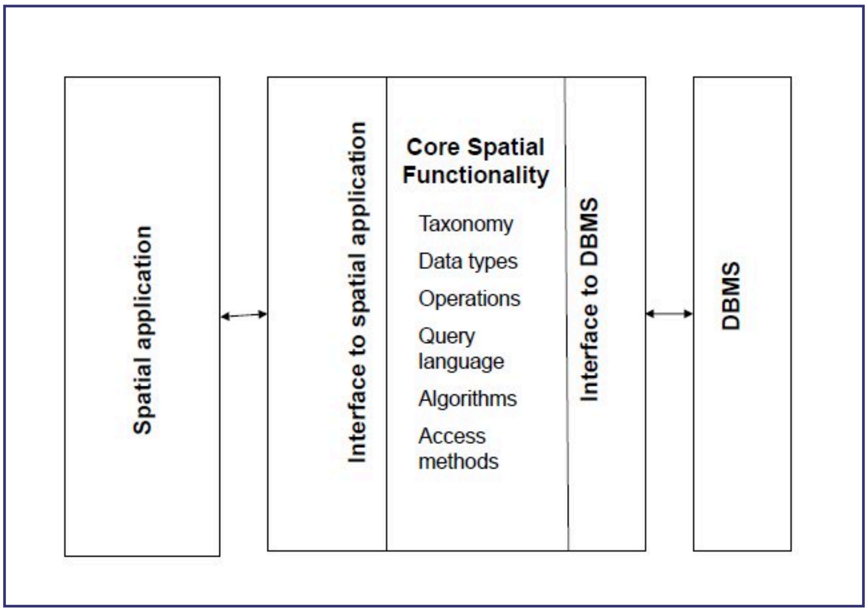


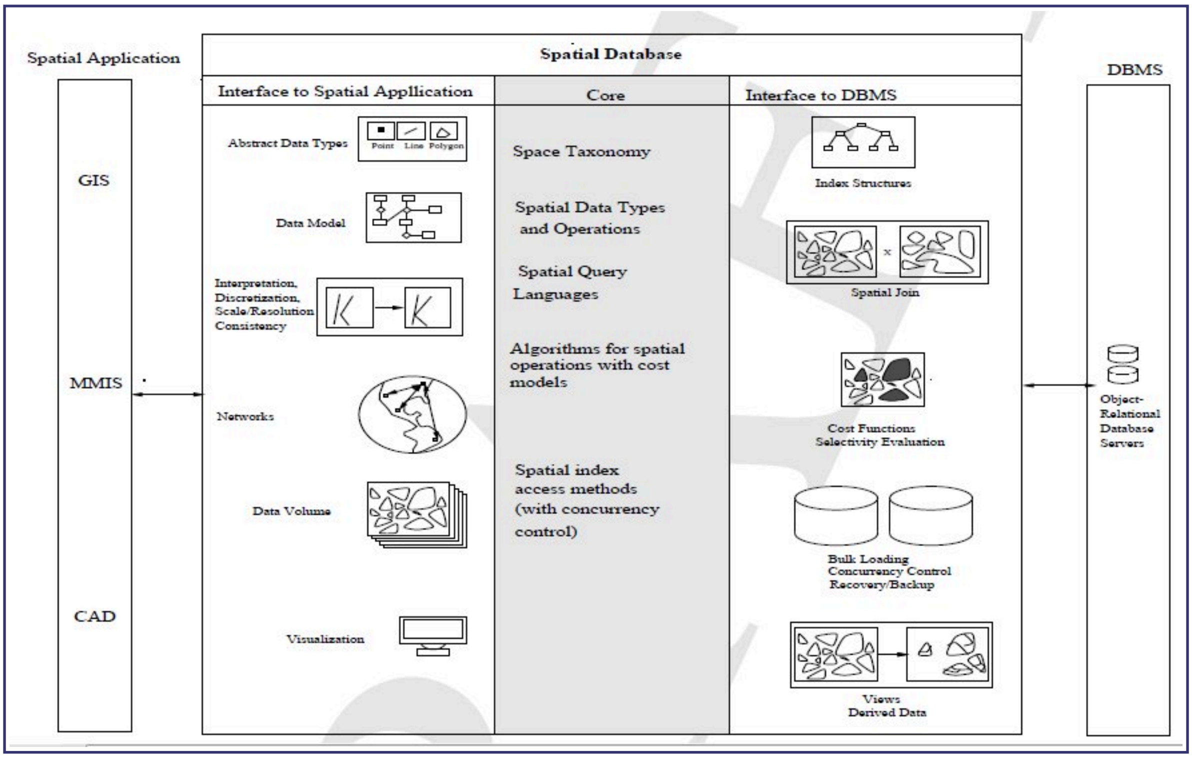
So a spatial DB is a collection of the following aspects, specifically built to handle spatial data:

- types
- operators
- indices

Soon, we will explore what types, operators and indices mean.

Spatial applications necessarily include spatial DBs:





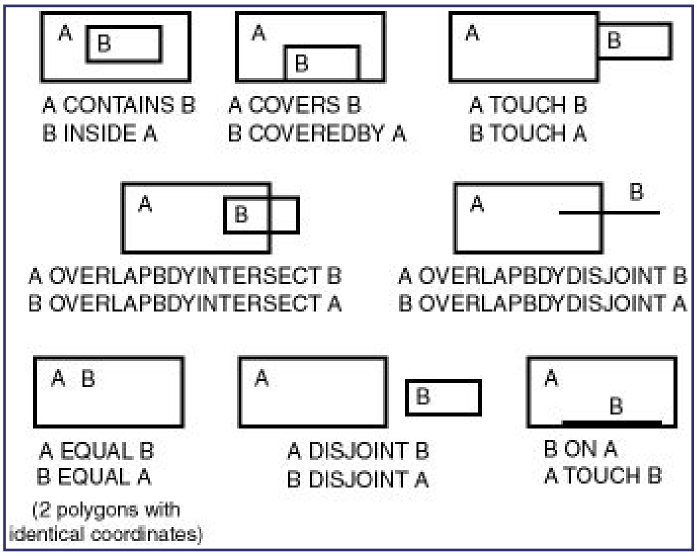
'GIS' is a specific application architecture built on top of a [more general purpose] SDBMS. GIS typically tends to be used for analyses that involve aspects of our (Earth) geography:

Search	Thematic search, search by region, (re-)classification
Location analysis	Buffer, corridor, overlay
Terrain analysis	Slope/aspect, catchment, drainage network
Flow analysis	Connectivity, shortest path
Distribution	Change detection, proximity, nearest neighbor
Spatial analysis/Statistics	Pattern, centrality, autocorrelation, indices of similarity, topology: hole description
Measurements	Distance, perimeter, shape, adjacency, direction

Spatial relationships

In 1D (and higher), spatial relationships can be expressed using 'intersects', 'crosses', 'within', 'touches' (these are T/F predicates).

Here is a sampling of spatial relationships in 2D:



Another diagram showing the [binary] operations:

IEEE Transactions on Knowledge and Data Engineering 6 (1): 86-95, 1994. 12

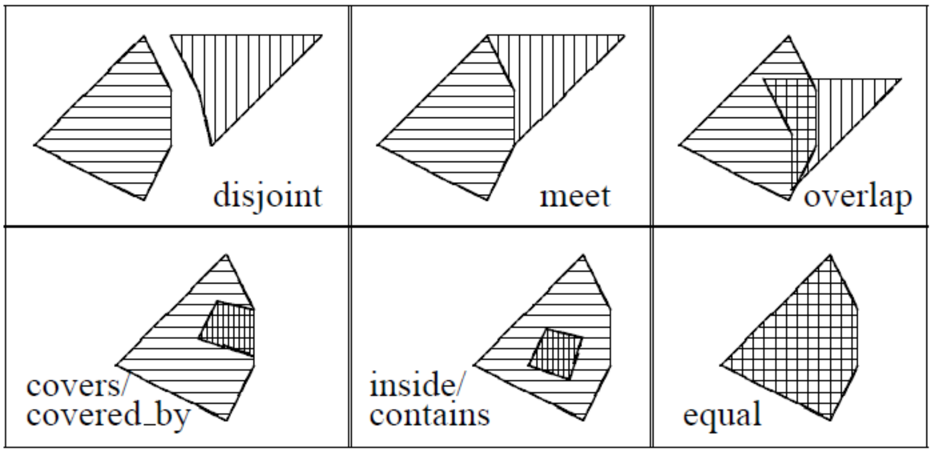
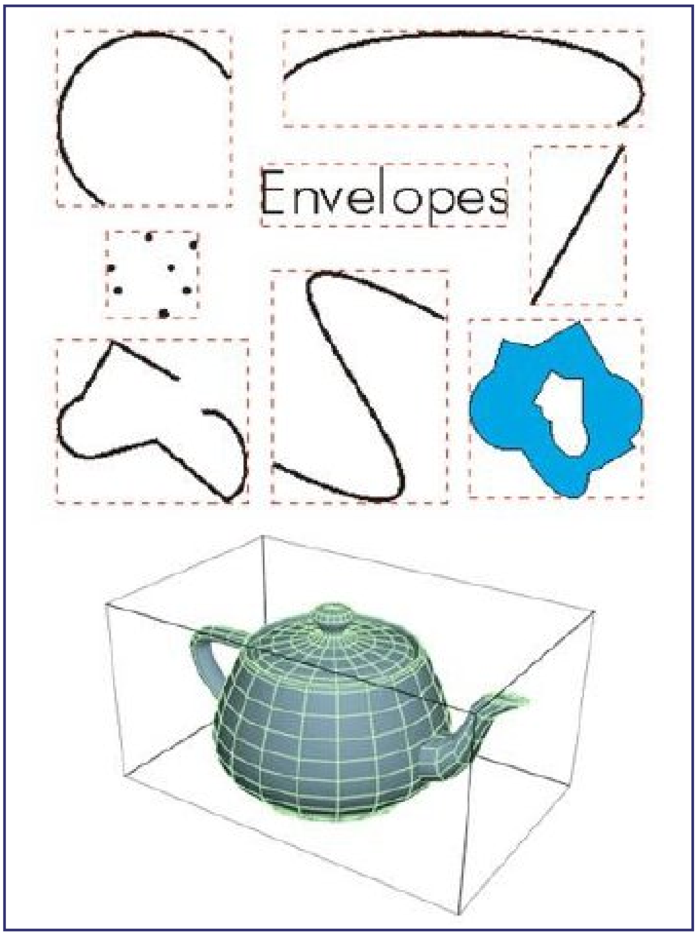


Figure 2. Examples of the binary topological relationships (a) disjoint, (b) meet, (c) overlap, (d) covers/covered_by, (e) inside/contains, and (f) equal [24].

Minimum Bounding Rectangles (MBRs) are what are used as a pre-processing ('filter') step, to compute the results of operations shown above:



Spatial relations - categories

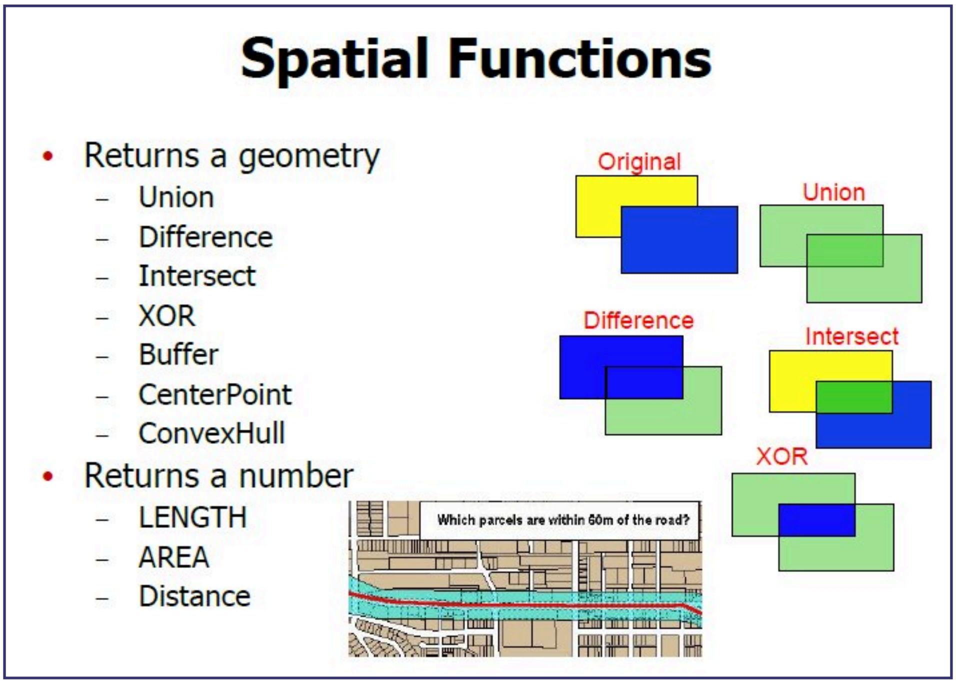
Spatial relationships can be:

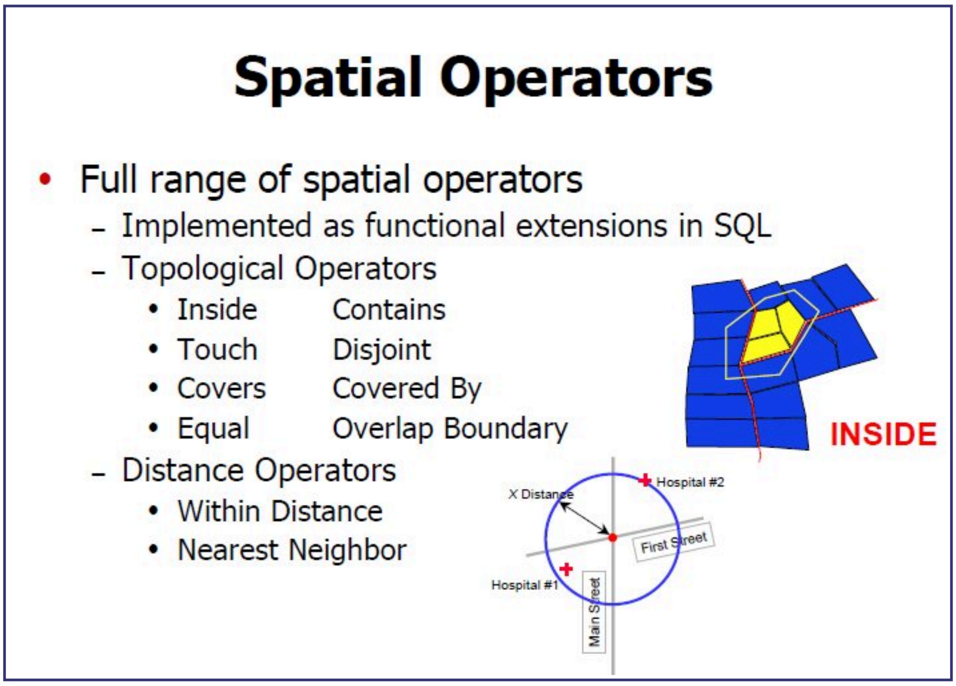
- topology-based [using defns of boundary, interior, exterior]
- metric-based [distance/Euclidian, angle measures]
- direction-based
- network-based [eg. shortest path]

Topological relationships could be further grouped like so:

- proximity
- overlap
- containment

Spatial operators, functions






```
#query

+ equals(another :Geometry) : Boolean
+ disjoint(another :Geometry) : Boolean
+ intersects(another :Geometry) : Boolean
+ touches(another :Geometry) : Boolean
+ crosses(another :Geometry) : Boolean
+ within(another :Geometry) : Boolean
+ contains(another :Geometry) : Boolean
...

#analysis

+ distance(another : Geometry) : Distance
+ buffer(another : Distance) : Geometry
+ convexHull() : Geometry
...
```

This doc provides more info on the various spatial operators.

How can we query spatial DBs? [how to 'call' the fns/ops]

We can perform the following operations, on spatial data:

- spatial measurements: find the distance between points, find polygon area..
- spatial functions: find nearest neighbors..
- spatial predicates: test for proximity, containment..

Spatial Data Entity Creation

- Form an entity to hold county names, states, populations, and geographies

```
CREATE TABLE County(
   Name       varchar(30),
   State      varchar(30),
   Pop        Integer,
   Shape      Polygon);
```

Spatial Data Entity Creation (Cont.)

- Form an entity to hold river names, sources, lengths, and geographies

```
CREATE TABLE River(
   Name       varchar(30),
   Source     varchar(30),
   Distance   Integer,
   Shape      LineString);
```

Example Spatial Query

- Find all the counties that border on Contra Costa county

```
SELECT    C1.Name
FROM      County C1, County C2
WHERE     Touch(C1.Shape, C2.Shape) = 1
          AND C2.Name = 'Contra Costa';
```

Example Spatial Query (Cont.)

- Find all the counties through which the Merced river runs

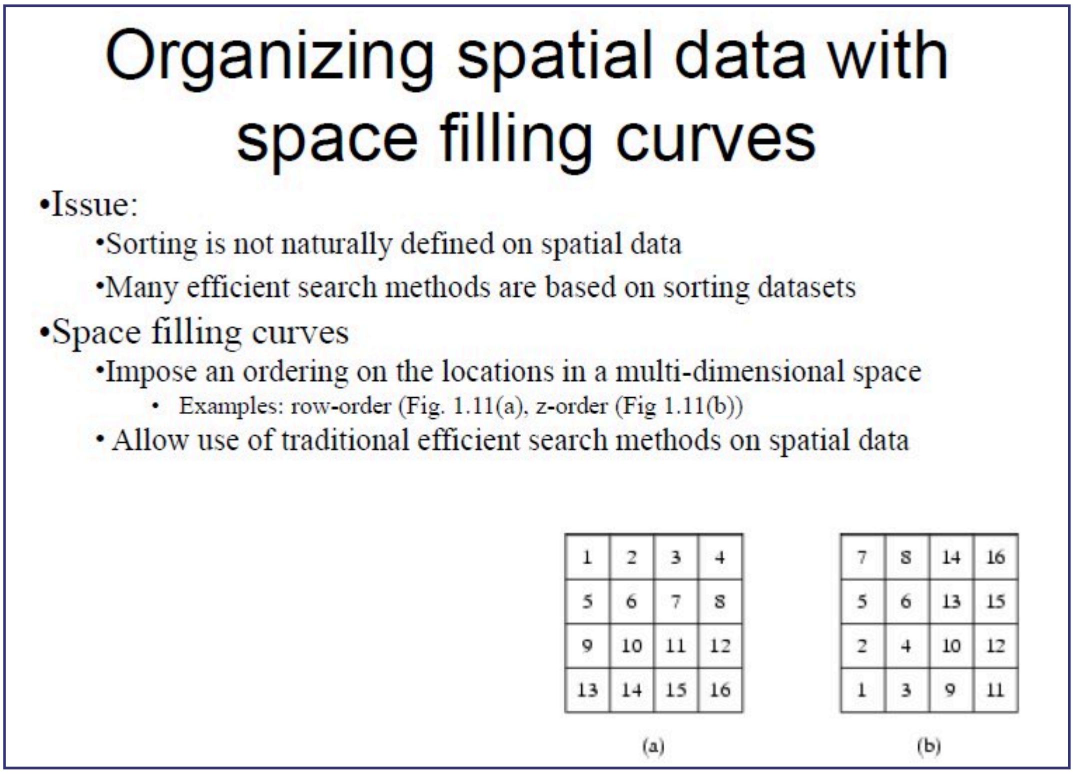
```
SELECT    C.Name, R.Name
FROM      County C, River R
WHERE     Intersect(C.Shape, R.Shape) = 1
          AND R.Name = 'Merced';
```

Creating spatial indexes

As (more so than) with non-spatial data, the creation and use of spatial indexes VASTLY speed up processing!

Can B Trees index spatial data?

In short, YES, if we pair it up with a 'z curve' indexing scheme (using a space-filling curve):

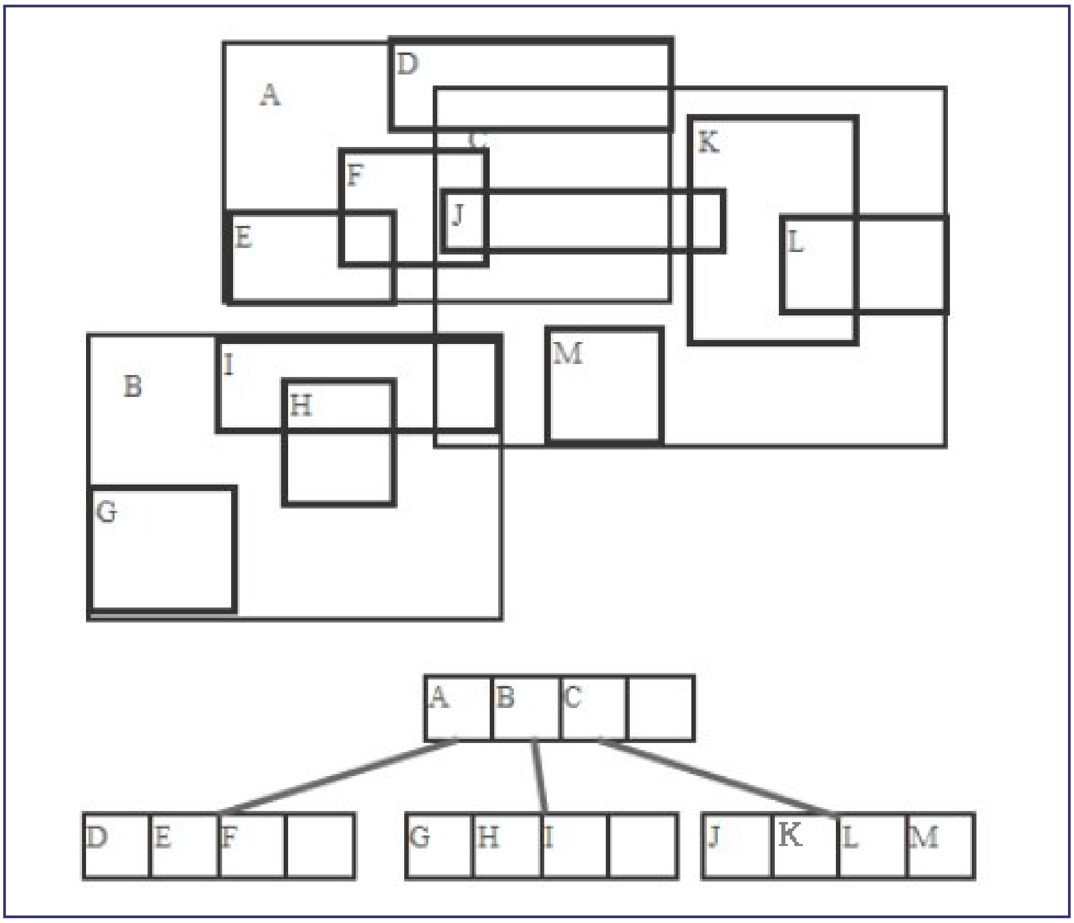


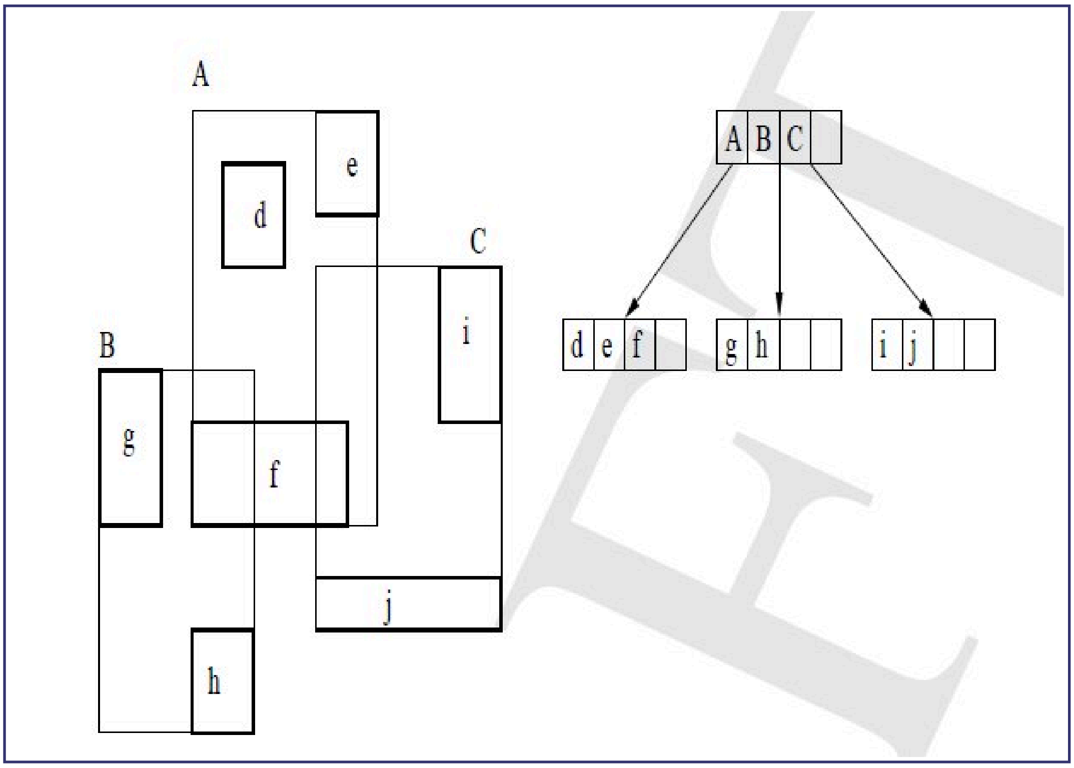
0:00 / 0:30

The idea is to quantize every (x,y) location into a recursively-divided 'quadtree' cell, and use the cell's binary (x,y) location to create a (binary) 'z' key, which is ordered along the unit (0..1) interval - in other words, 2D (x,y) points get mapped (indexed) to ordered 1D 'z' locations.

But, this is of academic interest mostly, not commonly practiced in industry - Apple's FoundationDB is an exception.

The 'R-tree' is the most common indexing data structure.





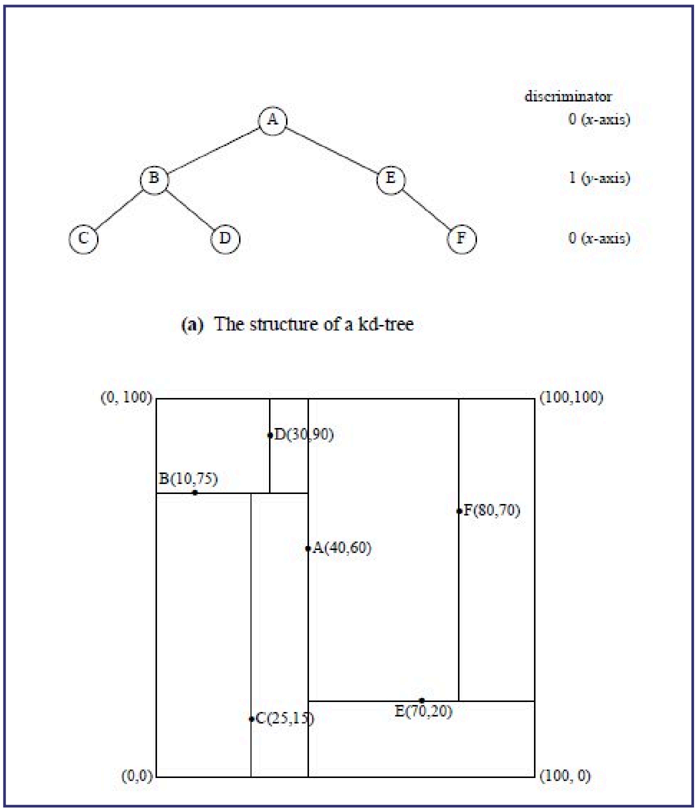
R trees use MBRs to create a hierarchy of bounds.

Variations, FYI: R+ tree, R* tree, Buddy trees, Packed R trees..

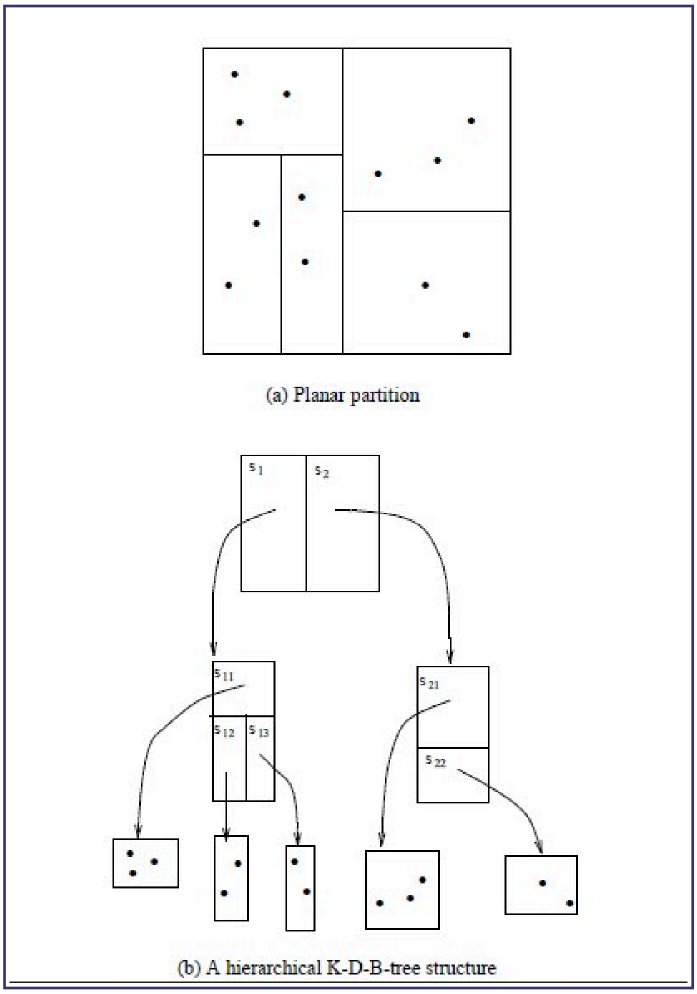
Also, here is an R tree for the entire world :)

R-tree alternatives

k-d tree:

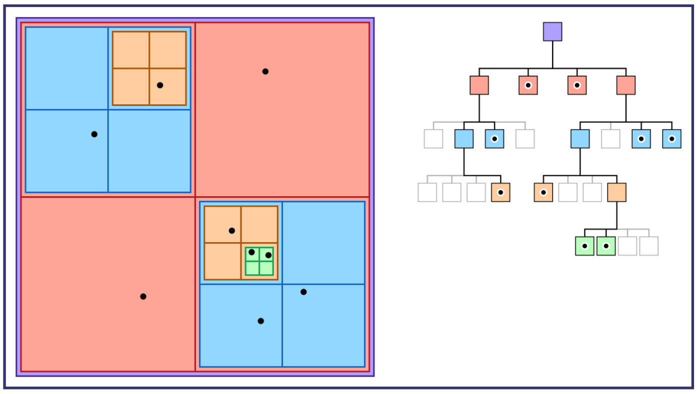


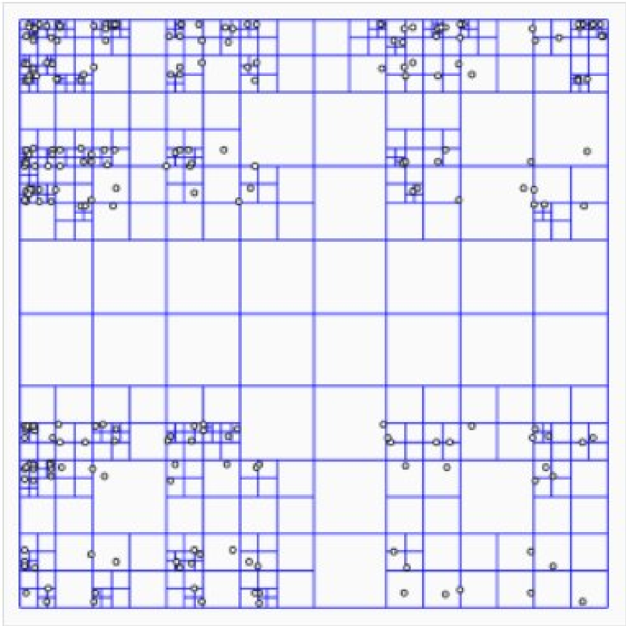
K-D-B tree:



Quadtree (octree in 3D):

Here we recursively and adaptively subdivide space [subdivisions happen only where necessary].

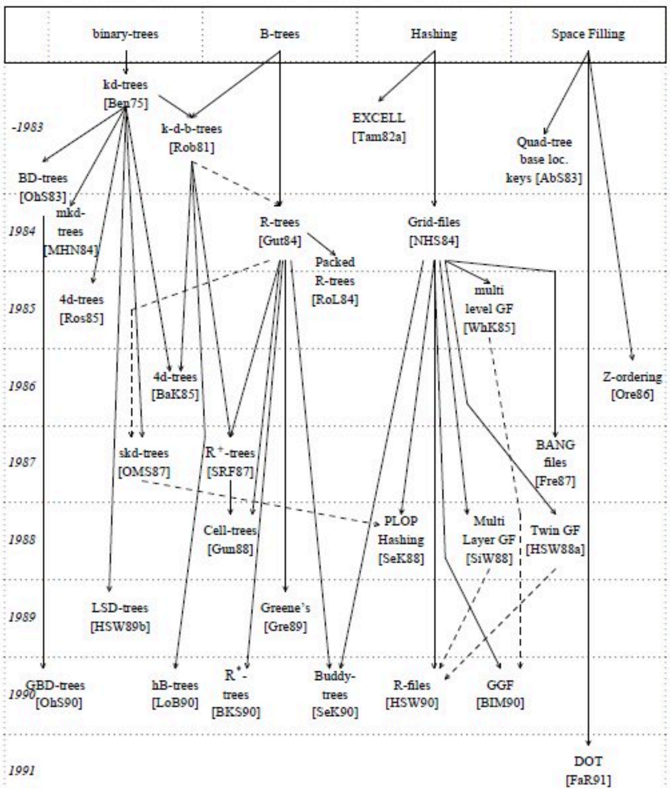




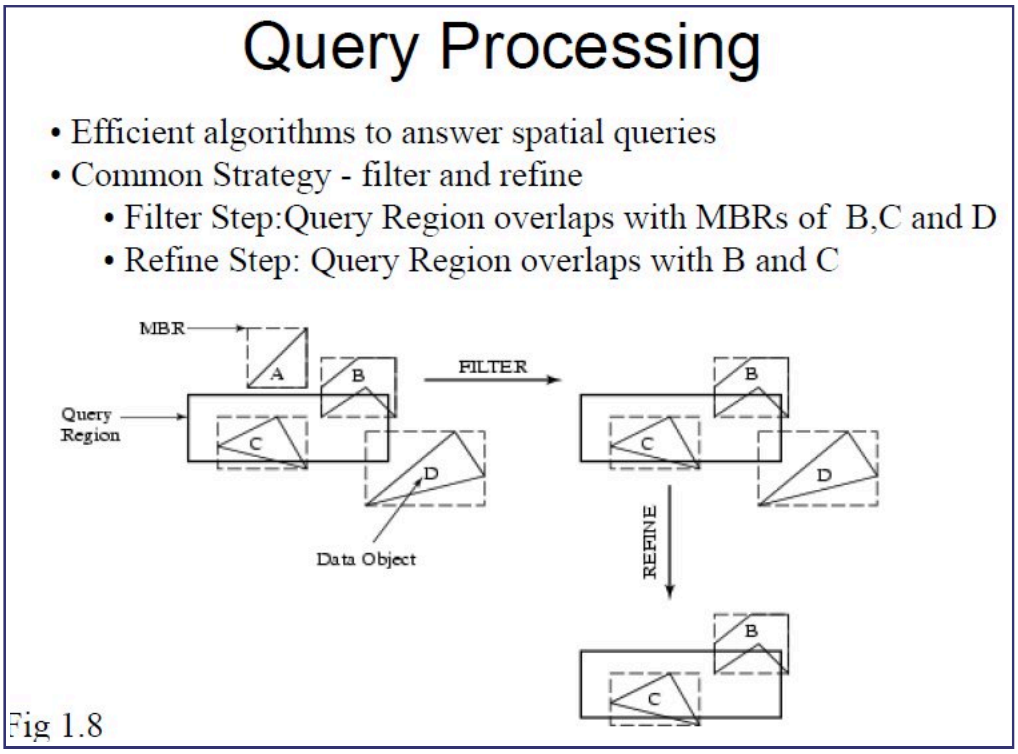
A representation of how a quadtree divides an indexed area. Source: Wikipedia

Each node is either a leaf node, with indexed points or null, or an internal (non-leaf) node that has exactly 4 children. The hierarchy of such nodes forms the quadtree.

Indexing schemes continue to evolve:

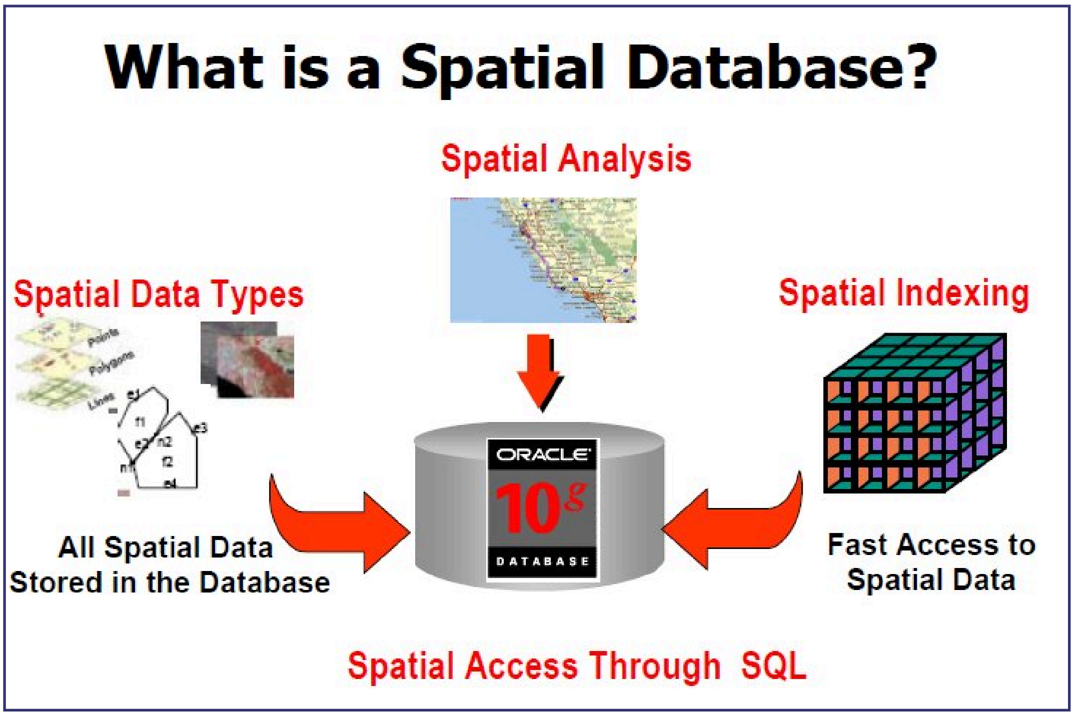


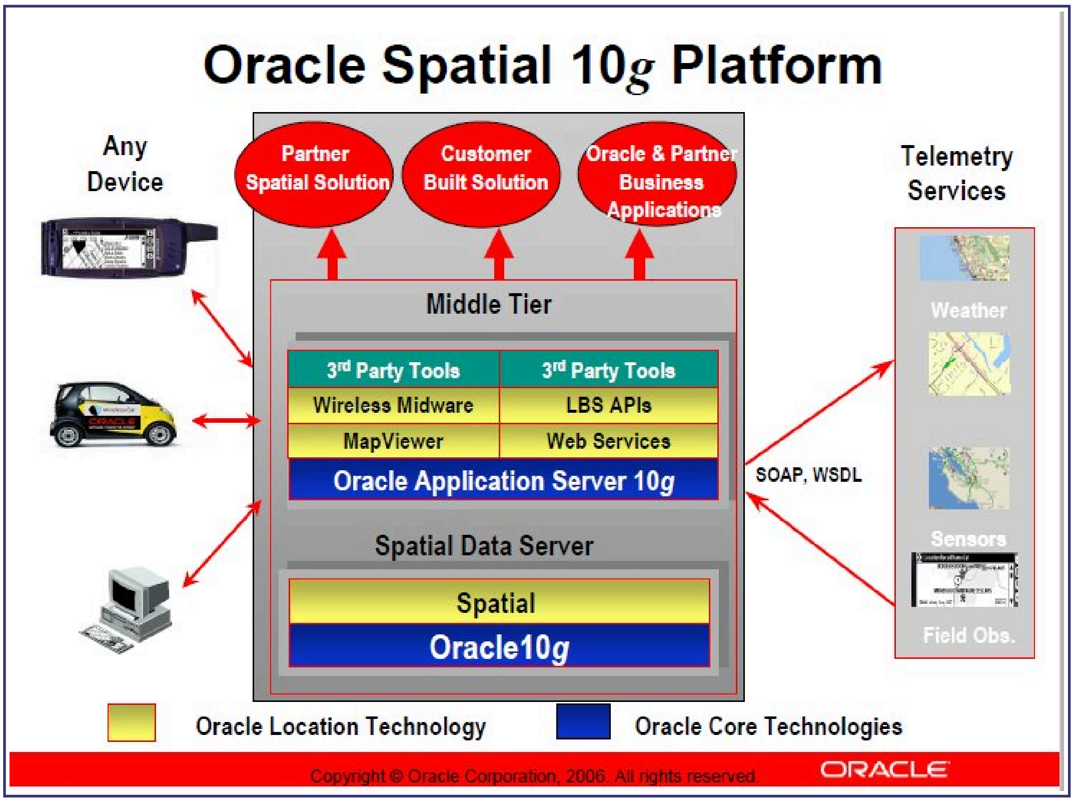
Query processing: filter, refine



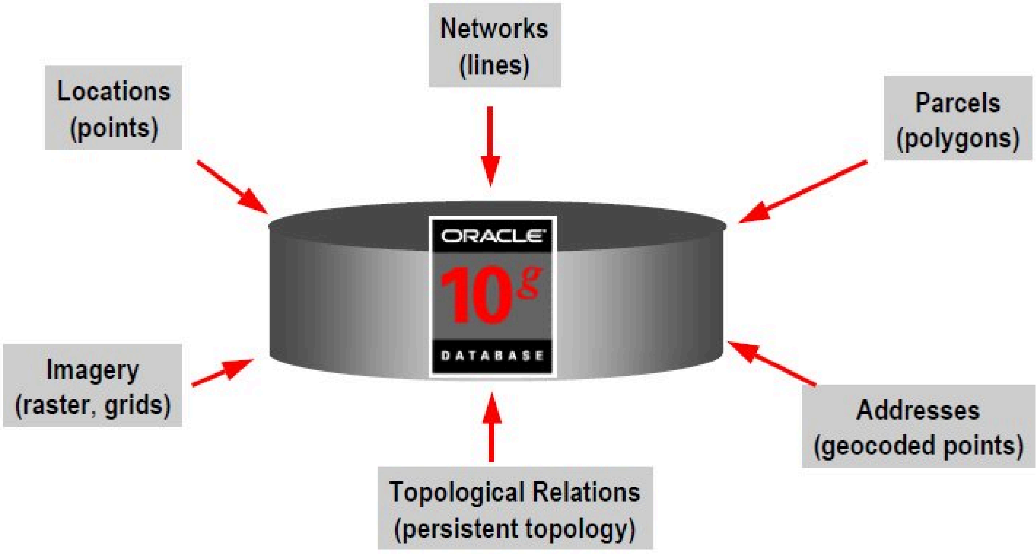
Spatial DB implementations (products)

Oracle offers a 'Spatial' library for spatial queries - this includes UDTs and custom functions to process them.





All Spatial Types in Oracle10*g*



SDO_GEOMETRY Object

- SDO_GEOMETRY Object

```
SDO_GTYPE           NUMBER
SDO_SRID            NUMBER
SDO_POINT           SDO_POINT_TYPE
SDO_ELEM_INFO       SDO_ELEM_INFO_ARRAY
SDO_ORDINATES       SDO_ORDINATE_ARRAY
```

- Example

```
SQL> CREATE TABLE states (
  2      state        VARCHAR2(30),
  3      totpop       NUMBER(9),
  4      geom         SDO_GEOMETRY);
```

SDO_GEOMETRY Object

- SDO_GTYPE - Defines the type of geometry stored in the object

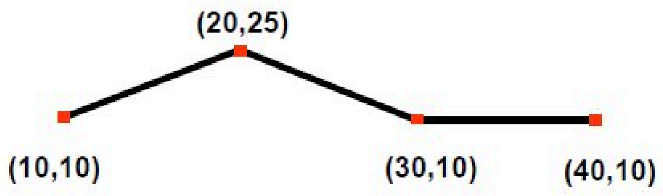
GTYPE	Explanation
1 POINT	Geometry contains one point
2 LINESTRING	Geometry contains one line string
3 POLYGON	Geometry contains one polygon
4 HETEROGENEOUS COLLECTION	Geometry is a collection of elements of different types: points, lines, polygons
5 MULTIPOINT	Geometry has multiple points
6 MULTILINESTRING	Geometry has multiple line strings
7 MULTIPOLYGON	Geometry has multiple polygons

SDO_GTYPE

SDO_GTYPE	Four digit GTYPEs - Include dimensionality		
	2D	3D	4D
1 POINT	2001	3001	4001
2 LINESTRING	2002	3002	4002
3 POLYGON	2003	3003	4003
4 COLLECTION	2004	3004	4004
5 MULTIPOINT	2005	3005	4005
6 MULTILINESTRING	2006	3006	4006
7 MULTIPOLYGON	2007	3007	4007

Constructing Geometries

```
SQL> INSERT INTO LINES VALUES (
  2>     attribute_1, …. attribute_n,
  3>     SDO_GEOMETRY (
  4>       2002, null, null,
  5>       SDO_ELEM_INFO_ARRAY (1,2,1),
  6>       SDO_ORDINATE_ARRAY (
  7>         10,10, 20,25, 30,10, 40,10))
  8>     );
```



Spatial Operators

- Operators
  - SDO_FILTER
    - Performs a primary filter only
  - SDO_RELATE and SDO_<relationship>
    - Performs a primary and secondary filter
  - SDO_WITHIN_DISTANCE
    - Generates a buffer around a geometry and performs a primary and optionally a secondary filter
  - SDO_NN
    - Returns nearest neighbors

SDO_FILTER Example

- Find all the cities in a selected rectangular area
- Result is approximate

```
SELECT c.city, c.pop90
FROM proj_cities c
WHERE sdo_filter (
    c.location,
    sdo_geometry (2003, 32775, null,
      sdo_elem_info_array (1,1003,3),
      sdo_ordinate_array (1720300,1805461,
                          1831559, 2207250))
    ) = 'TRUE';
```

Hint 1: All Spatial operators return `TRUE` or `FALSE`. When writing spatial queries always test with = `'TRUE'`, never <> `'FALSE'` or = `'true'`.

SDO_RELATE Example

- Find all counties in the state of New Hampshire

```
SELECT c.county, c.state_abrv
FROM geod_counties c,
     geod_states s
WHERE s.state = 'New Hampshire'
 AND sdo_relate (c.geom,
                 s.geom,
                 'mask=INSIDE+COVEREDBY')
                    ='TRUE';
```

Note: For optimal performance, don't forget to index `GEOD_STATES(state)`

Relationship Operators Example

- Find all the counties around Passaic county in New Jersey:

```
SELECT /*+ ordered */ a.county
FROM geod_counties b,
     geod_counties a
WHERE b.county = 'Passaic'
    AND b.state = 'New Jersey'
    AND SDO_TOUCH(a.geom,b.geom) = 'TRUE';
```

- Previously:

```
. . .
AND SDO_RELATE(a.geom,b.geom,
     'MASK=TOUCH') = 'TRUE';
```

SDO_NN Example

- Find the five cities nearest to Interstate I170, ordered by distance

```
SELECT /*+ ordered */
       c.city, c.state_abrv,
       sdo_nn_distance (1) distance_in_miles
FROM geod_interstates i,
     geod_cities c
WHERE i.highway = 'I170'
  AND sdo_nn(c.location, i.geom,
             'sdo_num_res=5 unit=mile', 1) = 'TRUE'
ORDER by distance_in_miles;
```

- Note: Make sure you have an index on `GEOD_INTERSTATES (HIGHWAY)`.

SDO_WITHIN_DISTANCE Examples

- Find all cities within a distance from an interstate

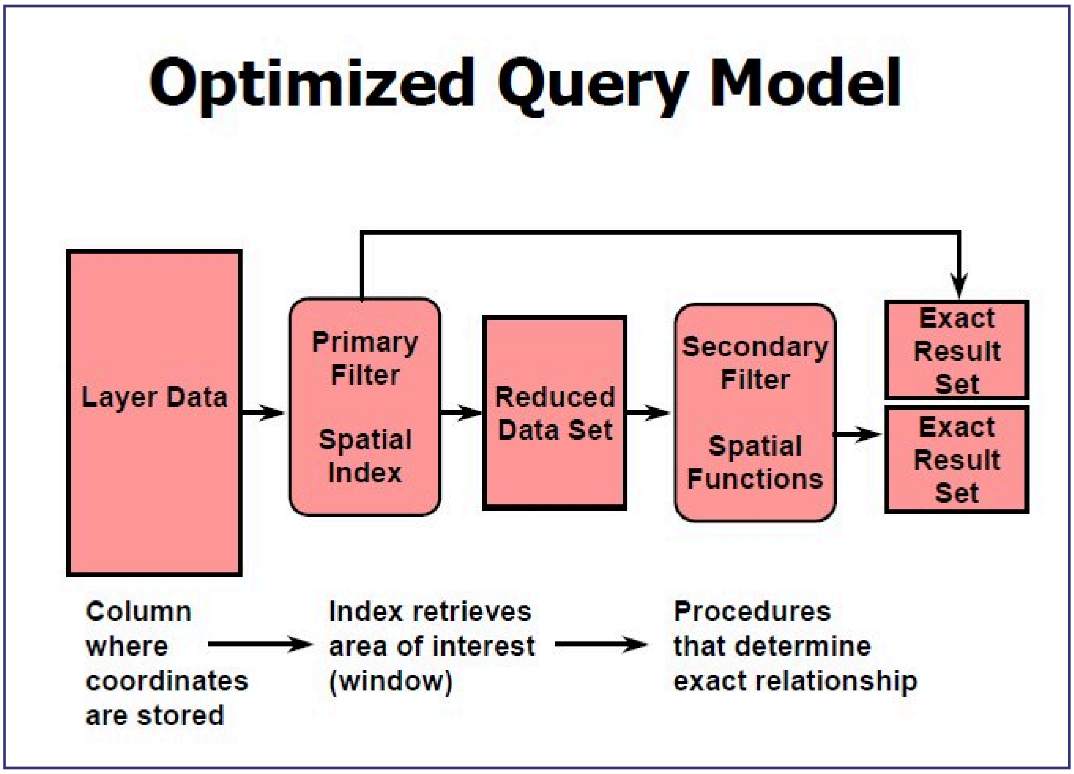
```
SELECT /*+ ordered */ c.city
FROM geod_interstates i, geod_cities c
WHERE i.highway = 'I170'
   AND sdo_within_distance (
        c.location, i.geom,
       'distance=15 unit=mile') = 'TRUE';
```

- Find interstates within a distance from a city

```
SELECT /*+ ordered */ i.highway
FROM geod_cities c, geod_interstates i
WHERE c.city = 'Tampa'
   AND sdo_within_distance (
        i.geom, c.location,
        'distance=15 unit=mile') = 'TRUE';
```

Spatial Indexing

- Used to optimize spatial query performance
- R-tree Indexing
  - Based on minimum bounding rectangles (MBRs) for 2D data or minimum bounding volumes (MBVs) for 3D data
  - Indexes two, three, or four dimensions
- Provides an exclusive and exhaustive coverage of spatial objects
- Indexes all elements within a geometry including points, lines, and polygons



Postgres' spatial DB library is called PostGIS.

Types of queries - PostGIS

The function names for queries differ across geodatabases. The following list contains commonly used functions built into PostGIS, a free geodatabase which is a PostgreSQL extension (the term 'geometry' refers to a point, line, box or other two or three dimensional shape):

Types of queries - PostGIS (Cont.)

1. Distance(geometry, geometry) : number
2. Equals(geometry, geometry) : boolean
3. Disjoint(geometry, geometry) : boolean
4. Intersects(geometry, geometry) : boolean
5. Touches(geometry, geometry) : boolean
6. Crosses(geometry, geometry) : boolean

Types of queries - PostGIS (Cont.)

7. Overlaps(geometry, geometry) : boolean
8. Contains(geometry, geometry) : boolean
9. Intersects(geometry, geometry) : boolean
10. Length(geometry) : number
11. Area(geometry) : number
12. Centroid(geometry) : geometry

Here is an example - table creation, and polygon insertion:

```
Create Table County (
name VARCHAR(30),
shape geometry);
CREATE TABLE


Insert into County values ('Lynn', ST_Polygon(ST_GeomFromText('LINESTRING(75.15 29.53 1,77 29 1,77.6 29.5 1,
 75.15 29.53 1)'),4326));
INSERT 0 1

SELECT *
FROM County;
 name |
         shape

------+------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------
-------------------
 Lynn | 01030000A0E6100000010000000400000009A99999999C9524048E17A14AE873D400000000000000F03F000000000040534000
0000000003D40000000000000F03F66666666666665340000000000000803D40000000000000F03F9A99999999C9524048E17A14AE873D
40000000000000F03F
(1 row)


Saty@Satys_USC_PC ~
$
```

You can learn a lot about PostGIS queries from this page.

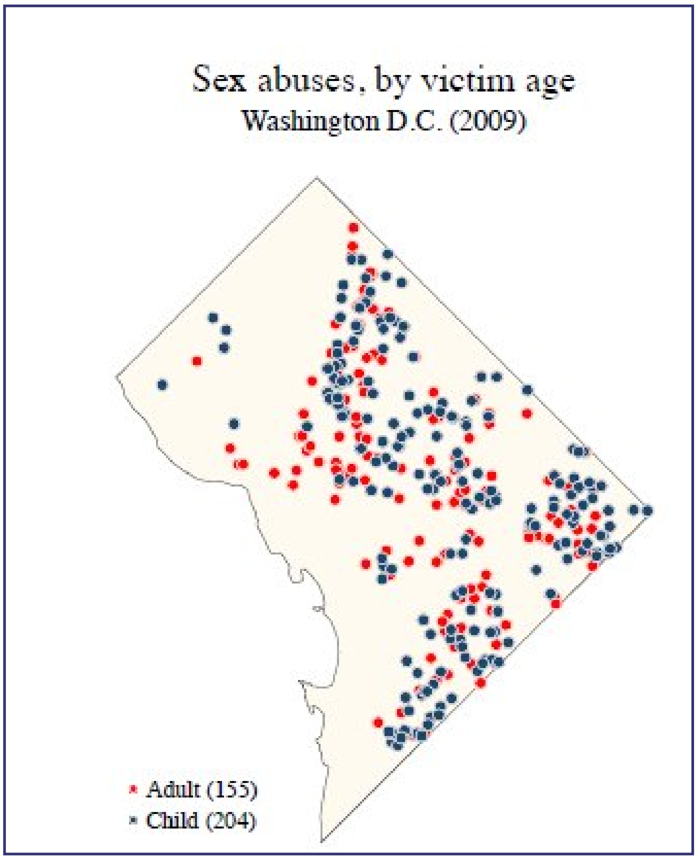
Thanks to SQL's facility for custom datatype ('UDT') and function creation ('functional extension'), "spatial" has been implemented for every major DB out there in addition to Oracle and Postgres:

- DB2: Spatial Datablade
- Informix: Geodetic Datablade
- SQL Server: Geometric and Geodetic Geography types
- MySQL: spatial library comes 'built in'
- SQLite: SpatiaLite
- ..

Visualizing spatial data

A variety of non-spatial attrs can be mapped on to spatial data, providing an intuitive grasp of patterns, trends and abnormalities. Following are some examples.

Dot map:



Here's another one.

Proportional symbol map:

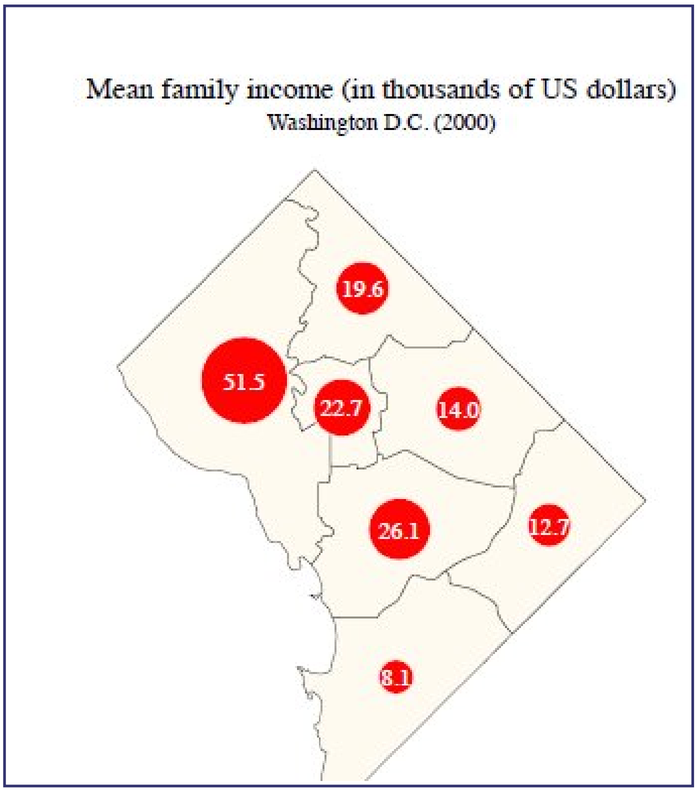
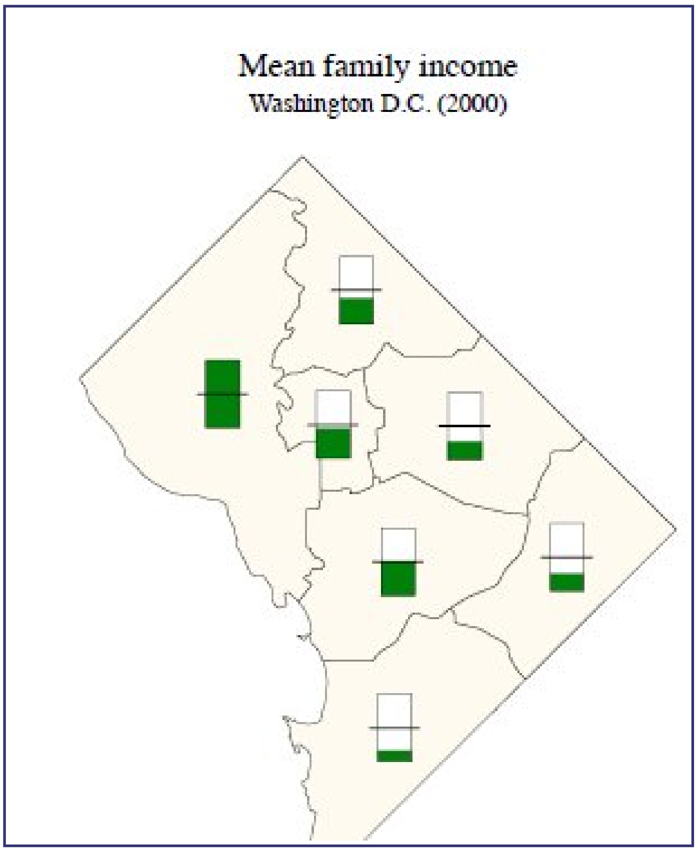
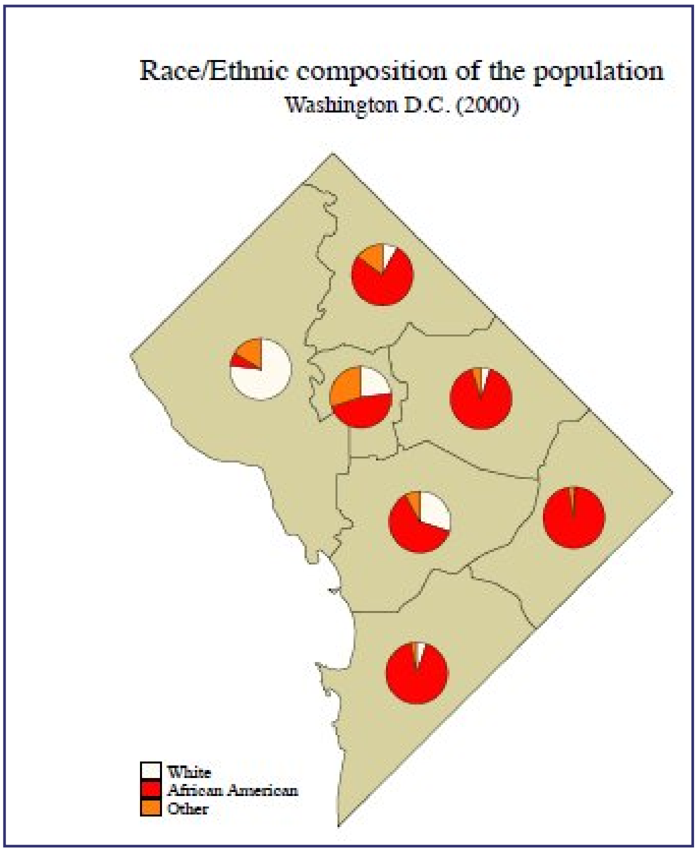


Diagram map:

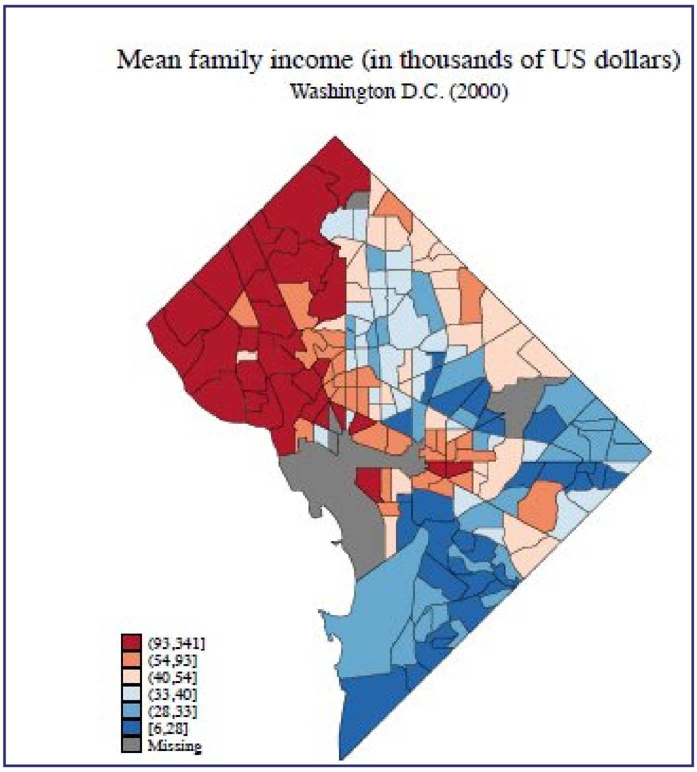


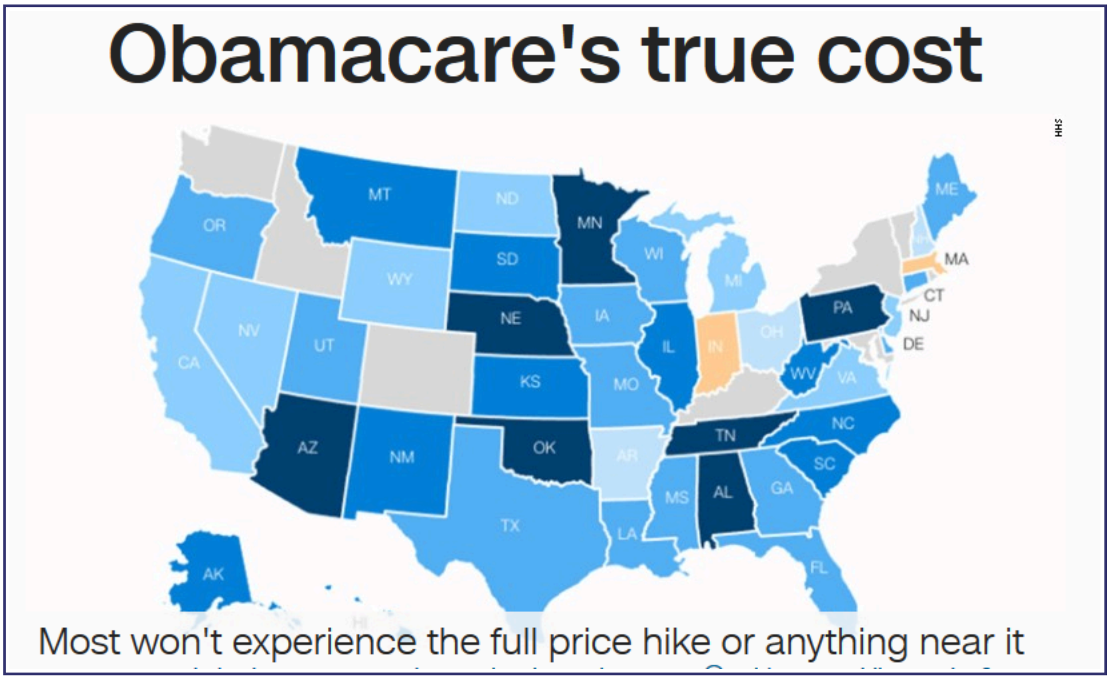
Another diagram map:



Also possible to plot multivariate data this way.

Choropleth maps (plotting of a variable of interest, to cover an entire region of a map):





Google KML

Google's KML format is used to encode spatial data for Google Earth, etc. Here is a page on importing other geospatial dataset formats into Google Earth.

More implementations

ESRI is the home of the powerful, flexible family of ArcGIS products - and they are local!

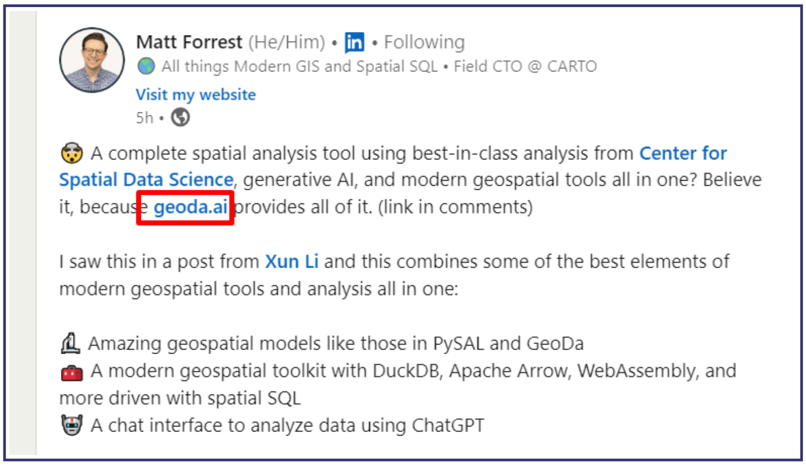
OpenLayers is an open GIS platform.

There is a variety of inexpensive/open source mapping platforms, competing with more pricey commercial offerings (from ESRI etc). Here are several:

- QGIS
- MapBox
- Carto
- GIS Cloud
- FME

Misc/MORE

https://www.uber.com/blog/h3/, https://www.uber.com/blog/visualizing-city-cores-with-h3/ [incl https://youtu.be/aCj-YVZ0mlE?t=200], p. 41/97 here.



Google's Gemini can handle geospatial data...

Nice: https://www.spatialintelligence.ai/p/the-untold-story-of-google-earth-5c4